

Ion channel gating

The written notes for the ion channel gating walkthrough.

To print: these notes as a two-column PDF.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

Leak channels and gated channels

- Leak channels have no gate. They are always open and let through whatever fits, all the time. Potassium leak channels are why a resting cell has a voltage at all.
- Most channels are gated: a moving part sits in the pore and blocks it until a specific trigger moves it.
- Gates let the cell control when ions move, not just whether they can. Leak channels set the resting voltage; gated channels let the cell act at a chosen moment.

Leak channel

A channel with no gate, open all the time.

Gated channel

A channel with a gate that opens only in response to a specific trigger.

What opens a gate

Channel	Opens when	Where you meet it
Voltage-gated	The voltage across the membrane changes. Charged parts of the channel sit in the electric field and physically move, and the gate moves with them.	The sodium and potassium channels of axons; calcium channels in axon terminals
Ligand-gated (chemically gated)	A specific molecule binds to a receptor site on the channel, like a key in a lock. The gate stays open while it is bound.	Acetylcholine receptors where a motor neuron meets a muscle fiber
Mechanically gated	The membrane is stretched, bent or pressed.	Sensory endings in the skin, hair cells of the inner ear, walls of blood vessels

- Positive feedback in voltage-gated sodium channels: channels open when the membrane becomes less negative, sodium comes in and makes it less negative still, which opens more channels. This runaway loop is why an action potential goes all the way once it starts.
- Ligand-gated channels are how one cell talks to the next: the message crosses the gap as a chemical, and the chemical opens gates on the far side.

- A drug that sits in the acetylcholine binding sites without opening the channels leaves the channels shut, and the muscle is not told to contract: weakness or paralysis. Some paralytic drugs used in surgery work this way. In myasthenia gravis, antibodies destroy these receptors, causing weakness that worsens with use.
- Mechanically gated channels are where sensation begins: pressure, sound vibration and stretch are turned into an electrical signal.

The voltage-gated sodium channel has two gates

- Activation gate: at the outer end of the pore. Closed at rest; opens quickly at threshold.
- Inactivation gate: at the inner end of the pore. Open at rest; responds to the same voltage change but moves slowly, and then plugs the pore from the inside.
- Both gates must be out of the way for sodium to pass.

State	Activation gate (outer)	Inactivation gate (inner)	Na ⁺ passes?	Can it open now?
Resting	Closed	Open	No	Yes, it is ready
Open	Open	Open	Yes, sodium floods in	It is open
Inactivated	Open	Closed, plugging the pore	No	No, whatever the stimulus

The sequence, every time the channel fires

1. At rest (about -70 mV) the channel is in the resting state.
2. The membrane depolarizes to threshold, about -55 mV, and the activation gate swings open.
3. Sodium pours in, driven by both its concentration gradient and the negative inside, so the voltage shoots up. This is the upstroke of the action potential.
4. About a millisecond later the slow inactivation gate closes and plugs the pore. Sodium stops, on its own, with nothing needed to switch it off.
5. The membrane repolarizes toward rest.
6. Only then does the channel reset: the activation gate closes, then the inactivation gate swings back out of the pore. The channel is back in the resting state.

Inactivation

The inner gate blocking the pore while the outer gate is still open. Different from being closed: a closed channel can open, an inactivated channel cannot.

Absolute refractory period

The time after firing when most sodium channels are inactivated, so no stimulus, however strong, can produce

another action potential. It sets the top firing rate and keeps an action potential traveling in one direction.

Channel reset

Return from inactivated to resting, which happens only when the membrane repolarizes.

The voltage-gated potassium channel

- It responds to the same depolarization as the sodium channel, but it has only one gate and it is slow.
- It is still opening while sodium channels are inactivating, so potassium leaves and helps bring the membrane back down.
- It closes late too, so potassium keeps leaving after the voltage is back at rest. The voltage dips below rest until the channels close.
- After-hyperpolarization: that dip below rest. It is part of why a neuron is harder to fire for a short time after an action potential, since the membrane is further from threshold.

Reading patients with the channel states

Patient	What is wrong	Why
Local anesthetic (lidocaine) at the dentist	Numb, but still able to move the face	Lidocaine sits in the sodium channel pore, so sodium cannot enter even when the activation gate opens. No action potential, no pain signal. It binds best to open or inactivated channels, so it acts fastest on nerves firing often, and small pain fibers are blocked before larger motor fibers.
Severe hyperkalemia	Excitable tissue stops responding; life threatening in the heart	High outside potassium holds the membrane depolarized, so sodium channels stay inactivated and never reset.
Hyperkalemic periodic paralysis	Episodes of weakness with high blood potassium, normal between episodes	An inherited muscle sodium channel that is slow to inactivate keeps the membrane depolarized; channels stay inactivated, so the muscle cannot fire. Too much depolarization means no activity, not more.